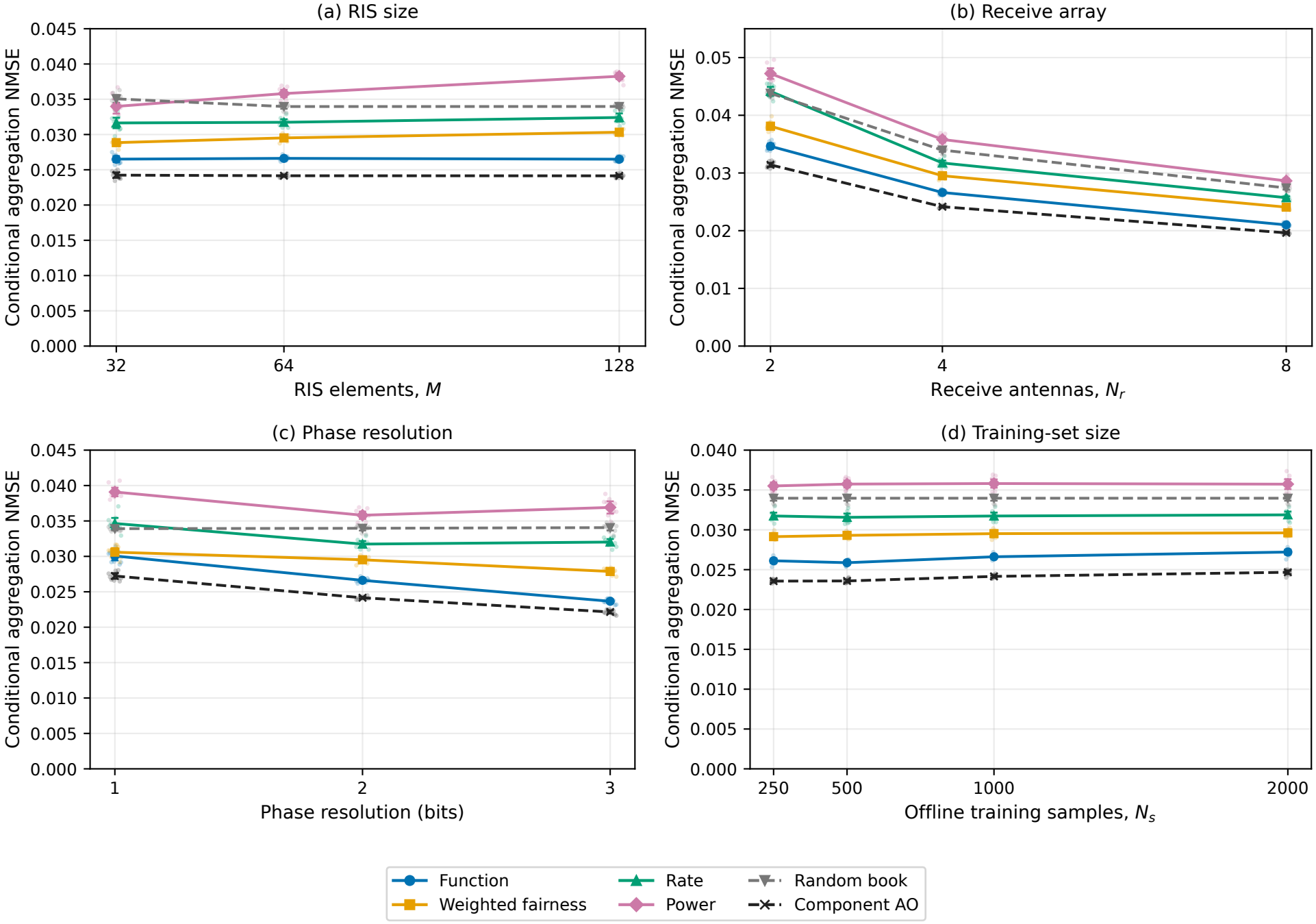

(a) RIS size
Conditional aggregation NMSE
0.045
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
32
64
128
RIS elements, M
(b) Receive array
Conditional aggregation NMSE
0.05
0.04
0.03
0.02
0.01
0.00
2
4
8
Receive antennas, N_r
(c) Phase resolution
Conditional aggregation NMSE
1
2
3
Phase resolution (bits)
(d) Training-set size
Conditional aggregation NMSE
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
250
500
1000
2000
Offline training samples, N_s
Function
Weighted fairness
Rate
Power
Random book
Component AO